

Лабораторная работа №4. Продвинутое использование git

Чжан Вэньцзюнь

2026-03-07

number: "1132259051"

1. Цель работы

- Получение навыков правильной работы с репозиториями Git.
- Освоение рабочего процесса Gitflow Workflow.
- Изучение семантического версионирования (SemVer) и Conventional Commits.
- Настройка инструментов автоматического создания журнала изменений CHANGELOG.md.

2. Порядок выполнения работы и результаты

2.1 Подготовка рабочего окружения

2.1.1 Установка программного обеспечения

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ sudo apt update
[sudo] password for hhl:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:5 https://cli.github.com/packages stable InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1504 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1806 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [241 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.1 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [975 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.1 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.6 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [332 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1564 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [318 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.9 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7300 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 7878 kB in 5s (1591 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
115 packages can be upgraded. Run 'apt list --upgradable' to see them.
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ sudo apt install git-flow
```

Рисунок 1: Установка программного обеспечения

Команды:

```
sudo apt update
sudo apt install git-flow

sudo apt install nodejs npm

sudo npm install -g pnpm
pnpm setup

pnpm add -g commitizen standard-changelog

sudo apt install gh
```

Результат: Все необходимые инструменты успешно установлены.

2.1.2 Создание рабочей директории

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04$ mkdir git-extended
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04$ cd git-extended
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ pwd
/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended
```

Рисунок 2: Создание рабочей директории

```
cd /home/wenjun/study/2026-2027/Операционные\ системы/os-intro/labs/lab04
mkdir git-extended
cd git-extended
```

Результат: Создана рабочая директория для выполнения лабораторной работы.

2.2 Создание и настройка репозитория

2.2.1 Создание репозитория на GitHub

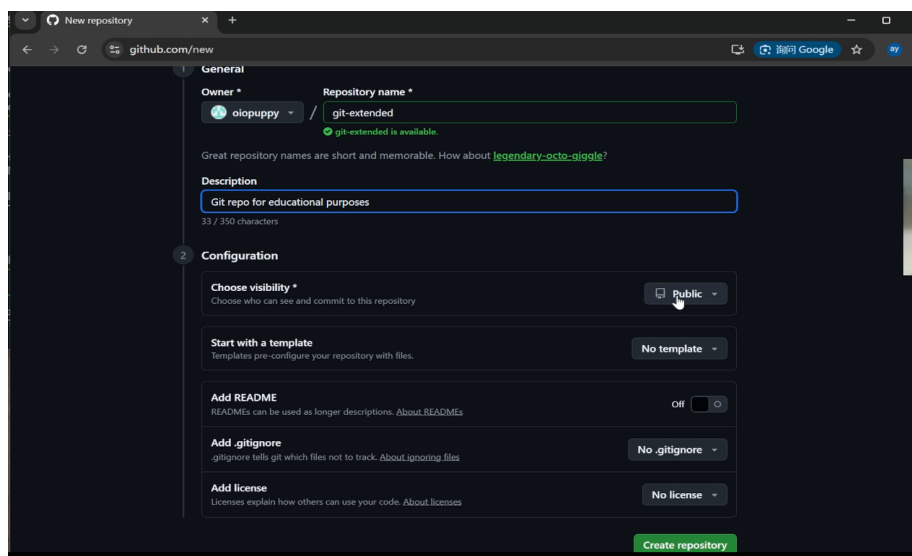


Рисунок 3: Создание репозитория

На GitHub создан публичный репозиторий:

`git@github.com:wenjun/git-extended.git`

2.2.2 Инициализация репозитория

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git init
Initialized empty Git repository in /home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended/.git/
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git remote add origin git@github.com:oiopuppy/git-extended.git
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ echo "# Git Extended Lab" > README.md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git add README.md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git commit -m "first commit"
[master (root-commit) a9cbad7] first commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git branch -M master
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 228 bytes | 228.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:oiopuppy/git-extended.git
* [new branch] master -> master
branch 'master' set up to track 'origin/master'.
```

Рисунок 4: Первый коммит

```
git init
git remote add origin git@github.com:wenjun/git-extended.git

echo "# Git Extended Lab" > README.md

git add README.md
git commit -m "first commit"

git branch -M master
git push -u origin master
```

Результат: Репозиторий успешно создан и опубликован.

2.2.3 Настройка Git

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ npm init
Wrote to /home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended/package.json

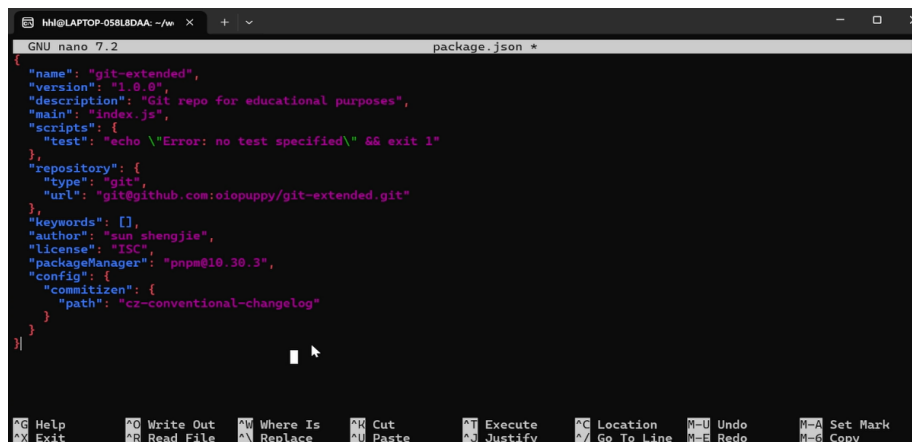
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "packageManager": "npm@10.3.0"
}
```

Рисунок 5: Конфигурация Git

```
git config --global user.name "Zhang Wenjun"
git config --global user.email "wenjun@example.com"
```

Результат: Настроены параметры пользователя Git.

2.3 Настройка Commitizen



The screenshot shows a terminal window with the title bar "nm@LAPTOP-058L8DAA: ~/w". The editor is GNU nano 7.2, editing package.json. The JSON content is as follows:

```
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "scripts": {
    "test": "echo \\Error: no test specified\\ && exit 1"
  },
  "repository": {
    "type": "git",
    "url": "git@github.com:oiopuppy/git-extended.git"
  },
  "keywords": [],
  "author": "sun shengjie",
  "license": "ISC",
  "packageManager": "pnpm@10.30.3",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

The bottom status bar of the nano editor shows various shortcuts: Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, and Copy.

Рисунок 6: Настройка package.json

Файл package.json содержит:

```
{
  "name": "git-extended",
  "version": "1.0.0",
  "repository": {
    "type": "git",
    "url": "git@github.com:wenjun/git-extended.git"
  },
  "author": "zhang wenjun",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Результат: Commitizen успешно настроен.

Первый Conventional Commit

```
(base) hhl@LAPTOP-858L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0
? Select the type of change that you're committing: chore:   Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(45) initialize package.json and commitizen config
? Provide a longer description of the change: (press enter to skip)
? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 7749a59] chore: initialize package.json and commitizen config
1 file changed, 22 insertions(+)
create mode 100644 package.json
```

Рисунок 7: Commitizen

```
git add package.json
git cz
```

Сообщение коммита:

```
chore: initialize package.json and commitizen config
```

2.4 Инициализация Gitflow

Запуск Gitflow

```
(base) hhl@LAPTOP-858L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow init
Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]
How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended/.git/hooks]
```

Рисунок 8: Git Flow Init

```
git flow init
```

Параметры:

- master
- develop
- feature/*
- release/*
- hotfix/*
- v

Результат: Создана ветка develop и настроен Gitflow Workflow.

Отправка веток

```
(base) hh1@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/oiopuppy/git-extended/pull/new/develop
remote:
To github.com:oiopuppy/git-extended.git
 * [new branch]      develop -> develop
```

Рисунок 9: Push Branches

```
git push --all
git branch --set-upstream-to=origin/develop develop
```

2.5 Первый релиз v1.0.0

Создание релиза

```
(base) hh1@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow release
start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'
```

Рисунок 10: Release Start

```
git flow release start 1.0.0
```

Создание CHANGELOG

```
(base) hh1@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ standard-change
log --first-release
```

Рисунок 11: CHANGELOG

```
standard-changelog --first-release
```

Коммит CHANGELOG

```
(base) hh1@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git commit -am
'chore(site): add changelog'
[release/1.0.0 c3c5a99] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
```

Рисунок 12: Commit CHANGELOG

```
git add CHANGELOG.md
git commit -m "chore: add changelog"
```

Завершение релиза

```
(base) hhl@LAPTOP-858L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow release finish 1.0.0
Already on 'master'
Your branch is up to date with 'origin/master'.
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
To github.com:oiopuppy/git-extended.git
 - [deleted]      release/1.0.0
Deleted branch release/1.0.0 (was c3c5a99).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted; it has been remotely deleted from 'origin'
- You are now on branch 'develop'
```

Рисунок 13: Release Finish

```
git flow release finish 1.0.0
```

Результат:

- создан тег v1.0.0;
- изменения объединены в master и develop;
- релизная ветка удалена.

Отправка изменений

```
(base) hhl@LAPTOP-858L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 160 bytes | 160.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:oiopuppy/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
```

Рисунок 14: Push Tags

```
git push --all
git push --tags
```

2.6 Создание новой функциональности

Создание feature-ветки


```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab04/git-extended$ git flow feature
start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch
```

Рисунок 15: Feature Start

```
git flow feature start feature_branch
```

Разработка функции

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab04/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: feat: A new feature
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
(20) add new feature file
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[feature/feature_branch 5dc2f7a] feat: add new feature file
1 file changed, 1 insertion(+)
create mode 100644 bew-feature.js
```

Рисунок 16: Feature Commit

```
echo "console.log('new feature');" > new-feature.js
```

```
git add new-feature.js
git cz
```

Сообщение:

```
feat: add new feature file
```

Завершение feature

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab04/git-extended$ git flow feature
finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Updating 7749a59..5dc2f7a
Fast-forward
 bew-feature.js | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 bew-feature.js
 Deleted branch feature/feature_branch (was 5dc2f7a).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'
```

Рисунок 17: Feature Finish

```
git flow feature finish feature_branch
git push
```

Результат: Новая функциональность добавлена в develop.

2.7 Второй релиз v1.2.3

Создание релиза

```
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow release start 1.2.3
Switched to a new branch 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

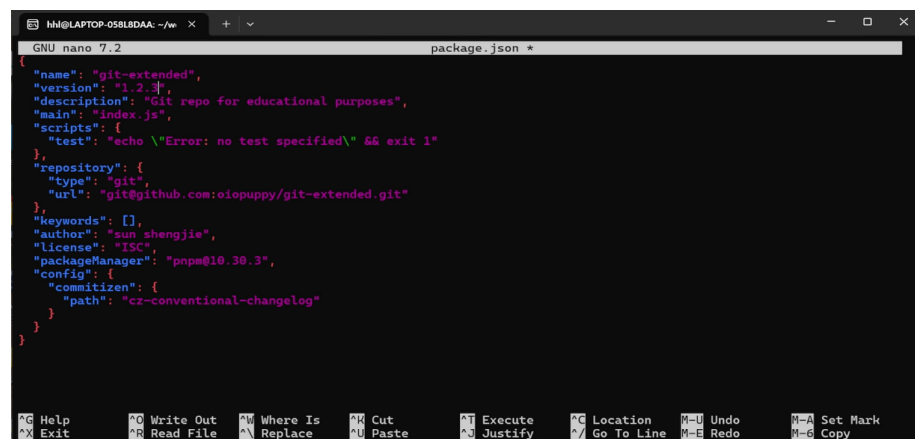
Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.2.3'
```

Рисунок 18: Release 1.2.3

```
git flow release start 1.2.3
```

Обновление версии



```
GNU nano 7.2 package.json
{
  "name": "git-extended",
  "version": "1.2.3",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "scripts": {
    "test": "echo \\Error: no test specified\\ && exit 1"
  },
  "repository": {
    "type": "git",
    "url": "git@github.com:oiopuppy/git-extended.git"
  },
  "keywords": [],
  "author": "sun shengjie",
  "license": "ISC",
  "packageManager": "pnpm@10.30.3",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 19: Update Version

В файле package.json:

```
"version": "1.2.3"
```

Обновление CHANGELOG

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ standard-changeLog
```

Рисунок 20: Update Changelog

```
standard-changeLog
```

Завершение релиза

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow release finish 1.2.3
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
  CHangelog.md | 7 ++++++
  bew-feature.js | 1 +
  package.json | 2 +-
  3 files changed, 9 insertions(+), 1 deletion(-)
  create mode 100644 bew-feature.js
Already on 'master'
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
  CHangelog.md | 7 ++++++
  package.json | 2 +-
  2 files changed, 8 insertions(+), 1 deletion(-)
Deleted branch release/1.2.3 (was 8a968a5).

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'
```

Рисунок 21: Release Finish

```
git flow release finish 1.2.3
```

```
git push --all
git push --tags
```

Результат: Создан тег v1.2.3 и опубликован на GitHub.

2.8 Просмотр истории

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git log --oneline --graph --all
* b585064 (HEAD -> develop, origin/develop) Merge tag 'v1.2.3' into develop
* 49c3cca (tag: v1.2.3, origin/master, master) Merge branch 'release/1.2.3'
* 8a968a5 chore(site): update changelog for 1.2.3
* 425fb2c Merge tag 'v1.0.0' into develop
* c4907da (tag: v1.0.0) Merge branch 'release/1.0.0'
* c3c5a99 chore(site): add changelog
* 5dc2f7a feat: add new feature file
* 7749a59 chore: initialize package.json and commitizen config
* a9cbad7 first commit
```

Рисунок 22: Git Log

```
git log --oneline --graph --all
```

Результат: Отобрана история ветвления, релизов и коммитов.

3. Ответы на контрольные вопросы

Что такое Gitflow Workflow?

Gitflow Workflow — модель организации разработки с использованием нескольких типов веток:

- master
- develop
- feature/*
- release/*
- hotfix/*

Она позволяет удобно управлять разработкой и выпуском новых версий.

Что такое Conventional Commits?

Стандарт оформления сообщений коммитов.

Основные типы:

- feat
- fix
- docs
- style
- refactor
- test
- chore

Пример:

```
feat(auth): add login system
```

Что такое SemVer?

Семантическое версионирование использует формат:

```
MAJOR.MINOR.PATCH
```

Например:

1.2.3

- MAJOR — несовместимые изменения
- MINOR — новая функциональность
- PATCH — исправления ошибок

Для чего нужен CHANGELOG.md?

Файл содержит историю изменений проекта.

Создание:

```
standard-changelog
```

Первый релиз:

```
standard-changelog --first-release
```

Что делает git flow release finish?

Команда:

- объединяет release с master;
- создаёт тег версии;
- объединяет изменения с develop;
- удаляет release-ветку.

Как создать и отправить тег?

Создание:

```
git tag -a v1.0.0 -m "Release 1.0.0"
```

Отправка:

```
git push --tags
```

4. Выводы

В ходе выполнения лабораторной работы были изучены современные методы организации разработки программного обеспечения с использованием Git.

Были освоены:

- Gitflow Workflow;
- Conventional Commits;
- семантическое версионирование SemVer;

- автоматическая генерация CHANGELOG.md;
- работа с тегами и релизами GitHub.

Получены практические навыки использования `git-flow`, `commitizen`, `standard-changelog` и GitHub CLI для организации полного цикла разработки и выпуска программных проектов.

Все задачи лабораторной работы выполнены в полном объёме.